The Hong Kong Polytechnic University
Department of Building Environment and Energy Engineering

BSE3227 HVACR 1 **2026/27 Semester 1**

Mini-design assignment

Students should form groups of up to five members, with each group tasked with designing the air conditioning layout for an open-plan office on a typical floor. The architectural layout plan for this floor is provided. It is assumed that a variable air volume (VAV) system will be utilized in the design.

Each group is required to complete the following tasks.

1. Space air diffusion design for maximum load condition

    - Calculate the cooling load of the open plan office using rule of thumb.
    - Determine the total supply air flow rate of the office.
    - Distribute supply air diffusers on the layout plan.
    - Calculate the supply air flow rate of each diffuser.
    - Find out the characteristic room length, L.
    - Refer to the performance data of KRUEGER "Louvered Face Diffusers 1400, 51400 Series 4-cone", and select a suitable type of diffuser.
    - From the diffuser performance data, find the corresponding $T_{0.25}$.
    - Calculate $T_{0.25}$ / L.
    - Check if it meets the ASHRAE ADPI requirement.

2. Space air diffusion design for minimum load condition

    - Calculate the supply air flow rate of each diffuser with a minimum turn down ratio of 40%.
    - From the diffuser performance data, find the corresponding $T_{0.25}$.
    - Calculate $T_{0.25}$ / L.
    - Check if it meets the ASHRAE ADPI requirement.

3. Single-line diagram of AC layout plan

    - Use single lines to sketch the air duct layout for distribution of supply air to all supply air diffusers in the office.
    - Use VAV boxes with four to five outlets, but limit the flexible duct length (from the VAV box to the diffuser) to be within 5 m.
    - Use straight ductwork as far as possible.
    - Minimize bends and turns.
    - Plan branch ducts to extend from the main trunk to distribute air evenly throughout the space.

4. Sizing of supply air ducts

- Determine the size of the main air ducts and each section of the branch air ducts.
- The air duct size should satisfy the requirements of maximum air velocity and limiting air-pressure drops.
- Use the Friction Chart in ASHRAE Fundamentals or Online Ductular.
- The maximum aspect ratio of rectangular air duct is preferred to be 4:1.
- Label each section of the air duct and tabulate the air duct sizing calculation in a table.
- The following table for air duct sizing is suggested for your use.

Section	Volume flow rate (L/s)	Rectangular duct size (mm×mm)	Air velocity (m/s)	Pressure drop (Pa/m)